Xiaofeng Li

AI Undergraduate · LLM Evaluation & Multimodal Perception

Guangzhou, China · xli683@connect.hkust-gz.edu.cn · 147 7150 6835 · c-allergic

Summary

I am Xiaofeng Li, a final-year undergraduate student in Artificial Intelligence at HKUST(GZ), graduating in 2027. My research focuses on large language model evaluation and multimodal perception. I am currently developing HiEval, an agent-based caption evaluation system, and I built IPV-Bench, the first benchmark for image protection methods in image-to-video generation, as first author (under review at AAAI 2027).

Education

BSc **The Hong Kong University of Science and Technology (Guangzhou)**, Artificial Intelligence — Guangzhou, China, Sept 2023 – June 2027

- GPA: 3.6+/4.3
- Dean's List, Spring 2025

Exchange Student **RWTH Aachen University**, Exchange Program — Aachen, Germany, Sept 2025 – Feb 2026

- One-semester study exchange in Germany

Experience

Caption Evaluation Agent System (HiEval), Developer — Apr 2026 – present, 6 months

- Built an LLM-agent-based caption evaluation system that hierarchically explores images from coarse to fine (region tree: ground → subdivide → decompose → match → commit → prune), verifying caption completeness against human-perception-aligned criteria and producing 3-D scores (P/R/F1, attributes, relations) with full decision traces — addressing the bias of surface-level element matching in existing benchmarks
- Automated the evaluation pipeline: produces fine-grained evaluation results and improvement suggestions without manual annotation, improving both efficiency and quality
- Overcame self-validation bias by building an independent grounding stack (GroundingDINO + SAM3 for objects, Mask2Former for background/stuff classes) that provides objective visual evidence
- Validated on four public benchmarks (CapsBench, CompreCap, CAPTURE, PerceptionRubrics) against 4 mainstream models (GPT-5.5, Claude Sonnet 4.5, Qwen3.5, Doubao Seed 2.1), evaluating 6600+ captions

Publications

IPV-Bench: Benchmarking Image Protection Methods under Diverse Image-to-Video Generation Scenarios — Mar 2026

The first systematic benchmark for image protection methods in image-to-video (I2V) generation scenarios — unifying evaluation protocols and generation settings, covering fidelity–effectiveness trade-offs, cross-model transfer, and robustness across diverse image domains, enabling reproducible and comparable evaluation.

Xiaofeng Li, Leyi Sheng, Yifan Zhao, Zhen Sun, Zongmin Zhang, Jiaheng Wei, Xinlei He

arxiv.org/abs/2603.26154 (arXiv:2603.26154 (Under review at AAAI 2027))

Programming Guide for Solving Constraint Satisfaction Problems with Tensor Networks Jan 2025

A practical programming guide demonstrating how to formulate and solve constraint satisfaction problems (e.g., graph coloring, SAT) with tensor-network-based methods, with reproducible Julia implementations.

Xuanzhao Gao, *Xiaofeng Li*, Jinguo Liu

arxiv.org/abs/2501.00227 (arXiv:2501.00227)

Awards

- Third Prize, Huawei Ascend Operator Challenge (S8), 2026
- Merit Award, Undergraduate Research Project Competition, HKUST(GZ), 2024
- Dean's List, HKUST(GZ), Spring 2025

Skills

Programming Languages: Python, Shell

Deep Learning & LLMs: PyTorch, Transformers, HuggingFace, LLM APIs & inference engineering

Computer Vision: GroundingDINO, SAM2/SAM3, Mask2Former

Tools: Git/GitHub, Claude Code, Linux, PowerPoint

Languages

- Chinese — Native
- English — IELTS 7.5